

When Does Portfolio Construction Work?

A Map across the Number of Assets and the Cost of Trading

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Abstract

A portfolio rule is a covariance estimate feeding an allocator. We ask a plain question: which rules actually work out of sample—after trading costs, and as the number of assets grows? We backtest eleven online constructions over thousands of random subsets of U.S. stocks and draw a map of the winner against two axes, the number of names and the cost of trading. The map has three regions. With few names and cheap trading, classic minimum-variance wins. With many names and cheap trading, a factor-model minimum-variance wins—and by a margin that grows with the number of names—because the ordinary version becomes undefined once there are more assets than data. Once trading costs rise above about five to ten basis points, simple inverse-variance weighting wins everywhere, because turnover, not estimation error, is what hurts. We then give the two ingredients that make the many-names corner work: a Woodbury inversion of a factor covariance, and a ridge-regularized version of the Schur coupling.

1 The question

Most studies compare portfolio methods one against another and crown a winner. A practitioner faces a different question: which method should I run, given how many names I hold, how noisy my covariance estimate is, and how much it costs me to trade? The answer is not a single method; it is a map.

Write a portfolio rule as

$$w = \mathcal{A}(\hat{\Sigma}) : \tag{1}$$

an allocator $\mathcal{A}$ (minimum-variance, risk parity, a tilt, ...) applied to an estimated covariance $\hat{\Sigma}$. Whether a rule is usable as the universe grows comes down to three things:

- **Is it defined?** With more assets than data, $\hat{\Sigma}$ is rank-deficient and $\hat{\Sigma}^{-1}$ does not exist. Any rule that needs the full inverse simply has no answer.
- **Is it fast enough?** An $O(n^3)$ step is unusable at thousands of names.
- **Does it pay after costs?** In-sample variance is not the point; net-of-cost return is.

This paper measures all three at once and reports the result as a two-axis map.

2 The methods, in one breath

We use online estimators: a one-time `fit`, then a cheap `partial_fit` each period over an exponentially-weighted covariance. This is the setting of online learning—one decision per period against a changing world, judged by regret (Cover 1991; Hazan 2016)—and the Thurstone tilt in particular is a follow-the-perturbed-leader rule, which perturbs abilities and takes the winner (Hannan 1957; Kalai and Vempala 2005). They split into two families by how they use $\hat{\Sigma}$.

Rules that never invert the covariance (work at any size). Equal weight; inverse-variance ($w_i \propto 1/\hat{\sigma}_i^2$, using only the diagonal); risk parity (equal risk contributions, an interior solution we solve by warm-started iteration) (Mailard, Roncalli, and Teiletche 2010); hierarchical risk parity (HRP), which allocates down a tree (Prado 2016); and the Thurstone tilt, which nudges a chosen benchmark and never touches Σ^{-1} .

Rules that invert the covariance (need it full-rank). Minimum-variance ($w \propto \Sigma^{-1}\mathbf{1}$); maximum diversification (Choueifaty and Coignard 2008); maximum decorrelation; and mean-variance. Each is a clean function of Σ , but undefined when Σ is singular—which is the normal state of affairs at scale.

The bridge. The Schur construction (Cotton 2024a) has a dial γ that slides from HRP ($\gamma=0$) to minimum-variance ($\gamma \rightarrow 1$).

Why turnover is built in, not bolted on. Because $w = \mathcal{A}(\hat{\Sigma})$, weights jolt around whenever either factor does. For the inverting rules the allocator is already a smooth function of Σ , so smooth weights just need a smooth covariance. The harder cases are HRP and the tilt, where the allocator itself is jumpy—a re-sorted tree, a fresh random draw—and we smooth those directly (a stable spectral ordering for the tree; a fixed random seed reused across updates for the tilt).

3 How we test

We use daily returns for the U.S. large-cap cross-section (Cotton 2024b); we redistribute no data. One trial works like this:

1. draw a random set of k names and a random time window;
2. walk forward—form weights from the past, earn the next day, update;
3. record gross Sharpe, turnover, and Sharpe after a c -bp trading cost.

We run thousands of trials with k and the window drawn at random. Averaging over trials removes the luck of which names and period a single backtest happened to pick—luck large enough, we found, to flip the ranking. One more convenience: net Sharpe is just gross minus a cost-proportional drag, so a single run at 10bp recovers the ranking at any cost.

4 What the map shows

4.1 At scale, trading cost decides—not skill

On broad subsets (hundreds of names) the gross and net winners are different methods. Minimum-variance has the best gross Sharpe, and its raw skill even grows with the number of names. But it trades a lot, and after costs it falls behind. Inverse-variance and risk parity trade roughly ten times less and win net. The crossover is early: about 5–10 bp.

4.2 The three regions

Table 1 is the headline: the winning method, by mean net Sharpe, over 2,638 trials. Read it in three pieces:

- **Few names, cheap trading $\Rightarrow$ minimum-variance.** The classic answer, while the covariance is still full-rank.

names $k \setminus$ cost	0 bp	5 bp	10 bp	20 bp
25	MinVar 0.59	MinVar 0.53	InvVar 0.50	InvVar 0.49
50	MinVar 0.71	MinVar 0.61	InvVar 0.56	InvVar 0.54
100	FactorMV 0.83	FactorMV 0.65	InvVar 0.54	InvVar 0.52
200	FactorMV 0.89	FactorMV 0.66	InvVar 0.56	InvVar 0.54
350	FactorMV 0.95	FactorMV 0.69	InvVar 0.56	InvVar 0.54
500	FactorMV 0.94	FactorMV 0.66	InvVar 0.53	InvVar 0.51

Table 1: The regime map: winner and its mean out-of-sample net Sharpe, by number of names and trading cost. Dense minimum-variance (MinVar) is undefined past the point where names exceed the data, so it is absent there; the factor-model version (FactorMV) takes that corner and wins by a widening margin (its win rate reaches 79% at $k=500$, zero cost).

- **Many names, cheap trading $\Rightarrow$ factor minimum-variance.** Here ordinary minimum-variance is undefined. The factor version (section 5) not only fills the gap, it wins outright, and its lead grows with the number of names.
- **Costs above ~ 10 bp $\Rightarrow$ inverse-variance,** everywhere. When trading is expensive, the rule that barely trades wins.

The novel coupling and tilt are competitive in the middle but rarely top the mean; their value is staying robust and well-defined at scale, not setting the net-Sharpe record.

4.3 How to set the Schur dial γ

Restricting to cases where the construction is well-defined, a middle setting of γ does beat HRP on raw skill—the best is near $\gamma \approx 0.25$ (0.539 versus HRP’s 0.529). But the edge is small and turnover climbs quickly with γ , so the moment trading costs anything the best γ drops to 0 (plain HRP) by about 2 bp. The rule of thumb: pick γ to maximise $\text{skill}(\gamma) - \text{cost} \times \text{turnover}(\gamma)$, which at realistic costs means keep it near zero.

4.4 Do the shorts pay? No.

The winning rules in the first two regions (minimum-variance and its factor version) allow short positions. Are those shorts earning their place, or are they just noise from inverting a noisy covariance? We test it directly: take the allocator’s own weight path, remove the shorts (clip to zero, renormalise),

and re-price it on the same returns. The shorts add value only if the signed book beats this long-only version out of sample. On every subset we tried, it does not.

- The short book grows with the number of names—its gross size reaches 180% at $k=400$ —which is the error-maximization one expects from an ill-conditioned inverse.
- Net of cost, dropping the shorts matches or beats keeping them at every dimension; only a sliver (+0.03 to +0.05 Sharpe) ever favours the shorts, and at large k they cost 0.2 or more.

So on this universe the negative weights are not needed: a long-only version of the same idea does as well or better after costs. This is consistent with the map, where the long-only inverse-variance rule already wins once trading is not free.

5 Making the many-names corner work

Two small fixes are what let the high-dimension column exist.

Factor minimum-variance (a stable inverse). Model the covariance as a few factors plus noise, $\Sigma = BB^\top + \text{diag}(\psi)$ with $B \in \mathbb{R}^{n \times k}$. Its inverse is then cheap and stable through the Woodbury identity,

$$\Sigma^{-1} = D^{-1} - D^{-1}B(I_k + B^\top D^{-1}B)^{-1}B^\top D^{-1}, \quad D = \text{diag}(\psi), \quad (2)$$

costing $O(nk^2)$. Keep the noise term ψ above a small floor and the inverse always exists, no matter how many names. That single change turns “minimum-variance is undefined” into the winner of table 1.

Ridge Schur (a stable coupling). The Schur dial needs to invert one block using another; at scale that other block is singular. Add a small ridge to that solve. It restores well-posedness and is self-correcting: when a block is poorly estimated the ridge dominates, the coupling fades, and Schur slides safely back toward HRP (a large ridge reproduces $\gamma=0$ exactly).

Scaling the tilt. The tilt’s naive cost is $O(Mn^2) + O(n^3)$. Writing its correlation in the same factor form makes a sample cost only $O(Mnk)$, so the benchmark-anchored, no-inverse tilt runs at about 0.4s per rebalance on 3,000 names.

6 What it all means

One idea ties the map together. In high dimension you can only reliably estimate simple structure: a diagonal, a few factors, or a tree. The methods that survive are exactly the ones that use only that—inverse-variance (diagonal), factor minimum-variance and the factor tilt (factors), HRP (tree). A rule that demands the full $n \times n$ inverse is trying to pin down $O(n^2)$ numbers from far less data; the only way to rescue it is to shrink it into one of those simple forms, at which point it has effectively become one of the survivors.

The second lesson is about cost. Because the cost axis decides the winner over so much of the map, an explicit turnover penalty and a constraint mechanism that does not jolt the weights are not niceties—they are what let a high-skill allocator keep its skill after costs. This sharpens the classic finding that naive diversification is hard to beat net of cost (DeMiguel, Garlappi, and Uppal 2009): the one place it is beaten is the cheap-trading, many-names corner, and the thing that beats it is a factor optimizer, not a dense one.

7 Conclusion

“Is this method good?” is the wrong question; “when is it good?” is the useful one. Across the number of names and the cost of trading, the answer is a map with three regions—dense minimum-variance, factor minimum-variance, and inverse-variance—divided by where the covariance runs out of rank and by a 5–10 bp cost line. At the frontier, the methods that work are the ones that never need a full inverse, made well-defined by a factor model and a ridge and made fast by a low-rank draw.

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